Multi-Source Candidate Data Transformer

Deterministic, evidence-first consolidation of recruiter CSVs, recruiter notes, and optional GitHub enrichment

Sources → Recruiter CSV + recruiter notes + GitHub profile URL | **Core guarantees** → deterministic, schema-validated, provenance-backed, confidence-scored
Configurable output → field selection, renaming, normalization, missing policy | **Failure behavior** → warnings for bad sources, no invented values, strict/partial modes

INPUTS & CONTROL		EVIDENCE		RESOLUTION		CANONICAL		OUTPUT
Manifest • CSV • Notes • GitHub Projection config	→	Adapters → normalize value • source • method • confidence	→	candidate_ref > GitHub > email contacts • fields • confidence	→	CanonicalCandidate + provenance JSON Schema gate	→	canonical or projected JSON warnings/errors • strict/partial

1 · Inputs become evidence

Manifest → Defines CSV files, note files, GitHub URLs, and optional candidate IDs. Relative paths resolve from manifest location.

Recruiter CSV → Assume structured ATS export (Workday/Eightfold-like). One row = one application record. Supported cells = observations: external candidate ID, name, contacts, job details, GitHub, application ID/time.

Recruiter notes → One file = one record. Extract email, phone, profile links, skills. Optional candidate ID links note to CSV candidate.

GitHub → One profile = one record. Extract name, email, bio, company, links, repository languages. API failure = warning; supplied URL still kept.

Normalize → Clean and validate values: lowercase email, E.164 phone, canonical URLs, dates, skills, company/title.

2 · Canonical output contract

Schema gate → Draft 2020-12. Required root fields. No unknown fields. Confidence always 0..1.

IDs → Input ATS ID becomes candidate_ref for matching/conflict checks; not returned as a profile field. candidate_id is always the generated cluster hash.

Candidate → name, location, links, headline, years_experience; all emails/phones + primary/secondary + details/status/reason/confidence; skills, experience, education.

Audit → Every accepted value keeps field, source, locator, method, confidence. Candidate also gets overall_confidence.

Current MVP → location, years_experience, education, experience.summary stay empty. Application ID/time used for resolution, not returned as profile fields.

CLI envelope → Written JSON contains status, output kind, candidate count, candidates, warnings, errors. Strict failed run writes no envelope.

3 · Runtime projection + validation

Output config → Per field: canonical source, output name/path, type, required flag, missing policy, optional normalizer. Optional confidence/provenance.

Preflight + project → Validate config before file/network work. Then read canonical field → expand index/list → rename/write → normalize → type-check.

Failure → Strict error: stderr + exit 1 + no JSON. --allow-partial: write survivors; zero survivors writes failed envelope, then exits 1.

4 · Deterministic merge + conflict policy

A. Decide whether records are the same candidate

- Match order → candidate_ref, GitHub, email. Same key can merge; name alone never merges.
- candidate_ref guard wins. Same email/GitHub + different refs → keep separate and warn. Unreferenced record stays neutral.
- Phone → merge only with another strong key or matching name tokens; group size ≤3.
- candidate_id → deterministic hash.

B. Resolve email and phone values

- Only CSV-backed contact can be primary because CSV is assumed structured ATS data. Notes/GitHub corroborate or remain secondary.
- Confidence → strongest observation + notes/application support; bonus ≤0.05, detail ≤0.99.
- Winner → one CSV value; else unique latest application; else unique highest confidence.
- Tie → ambiguous/no primary. No CSV → unstructured_only/no primary. Return all values.

C. Resolve remaining fields

- Scalars → confidence, longer value, source priority: CSV > notes > GitHub, then stable order.
- Skills → canonical name + source support.
- Experience → merge only when company/title/start/end match.

D. Confidence meaning

Observation → source × field × extraction × validation. Source factors: manifest 0.98, CSV 0.95, notes 0.80, GitHub 0.75.

Field factors → ref/CSV email 1.00; CSV name/phone 0.95; notes email 0.95, phone/link 0.90, skill 0.85; GitHub URL 1.00, email 0.90, profile fields 0.70–0.85.

Overall → 15% name + 25% email + 10% phone + 20% GitHub + 15% skills + 15% experience. Missing field = zero. Quality/completeness score, not merge probability.

5 · Edge cases + deliberate scope

- Conflicting IDs → block merge and warn.
- Equal contact evidence → ambiguous; never choose alphabetically.
- Local phone without region → reject value and warn.
- Negated skill / ambiguous “go” → skip.
- GitHub unavailable → keep URL, warn, continue.

Leave out → Fuzzy/LLM/name-only matching, resume/OCR parsing, database/review UI, real-time API, GitHub pagination/retry/cache/concurrency.